Automatic Text Summarization using Generative AI and Flask Web Application

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Abstract

The digital text information has tremendously increased within a very brief period and, therefore, the need to come up with efficient summarization software that would help to isolate substantial information on large volumes of documents. This paper will introduce an Automatic Text Summarization (ATS) web-based system, which is a combination of extractive and abstractive systems of summarizing to generate clear and concise summaries. The system proposed in the paper is founded on the principle of Natural Language Processing (NLP) systems and transformer-based systems of Generative Artificial Intelligence (AI) to provide clear and concise summaries of documents of various formats, i.e. plain text, PDF files, and PowerPoint files. The extractive part becomes used to select major sentences with the aid of statistical and relevancy based scores. In comparison, the abstractive part makes the large language models serve as summaries depending on the context and to simplify their readability and comprehension.The system is created in Python and Flask to create back-ends. It offers user friendly web interface which may be scaled and adjusted to allow the user to adjust the length of summaries and also, to up-load files. The conventional metrics, such as the ROUGE scores, give us an opportunity to evaluate the performance of this hybrid algorithm in terms of the content preservation and the language quality where, the experimental results testify to the fact that the hybrid algorithm is able to perform successful text compression without losing the context and its readability. This resolution shows the potential of Generative AI to enhance the study of papers and extracting knowledge in educational, business, and research settings.

Introduction

The quantity of text that has been produced on a daily basis has been increasing at a slow but steady pace due to the rising digital era. The unstructured texts information in the academic, online news, corporate, blogs and social media has been gradually and incessantly accumulating. This inundation of information, as much as it might include diffusion of knowledge, global communication in its true meaning, it has

presented a gigantic task of processing and making sense of this information. There is much time wastage before persons and organizations unearth some important information, identify important arguments, and make deductions. It is therefore transforming to a further level of intelligent automated software successfully condensing a huge amount of texts to brilliant and concise conclusions.

Automatic Text Summarization (ATS) is a sub area of the Natural Language Processing (NLP) which is preoccupied with the design of shorter texts and memorizing the central information, central themes and the meaning of the textual context without their corruption. The idea of ATS would be to perform the summations of the original which will be not only logical, but also informative and reflective of the original material in the absence of manipulation or exaggeration of the information. In most cases, overviewing information systems are needed to decrease the reading time, and grant the access to information in various and far-between locations: the research, business analytics, legal files, healthcare reporting, and educational material management.The convergence of text summarization procedures may be conventionally reduced to two tactics:

Extractive Summarization :-

Extractive processes of summarization process are processes which are capable of finding and selecting the largest sentences or phrases that they can locate in the original text. They are either dictated by statistics and lingual features analysis, rank decision algorithm on graphs, or a machine learning algorithm in order to provide sentence significance. The most widespread include Term Frequency-Inverse Document Frequency (TF-IDF), TextRank and supervised classification models. Admittedly, extractive methods are computationally inexpensive and not inaccurate about fact but are lower in the number of ways to be flexible and are more a summary of a text, which might appear to be discontinuous or more non-native.

Abstractive Summarization :-

Abstractive summarization is developed further. The abstractive models identify the general meaning of the document and generate new sentences, which present the important ideas, as opposed to dissection of the text where a sentence was included into it. The technique has been based on the dense artificial intelligence representations, sequence-to-sequence and transformer-based neural networks. Abstractive summarization is similar to human summarization i.e. it involves the rephrasing, sentence restructuring and inferences amongst information in various sections of a text. But it is time-violent regarding it computation power, and trained models.

The tremendous number of recent advances in Generative Artificial Intelligence (AI) and systems of abstractive summarization specifically have brought large strides to the output of the systems. Transformer based models such as BART, T5, and PEGASUS have attention mechanisms and can be applied to comprehend long-range shaped views and establish text contexts. These large language models are more informative using semantics and contextually sensitive in generation of similar constant, fluent and human like re-summaries. More so, these extensions have facilitated the use of the ATS systems to work with other documentations such as PDF files, research papers and presentation slides.

It was these developments that prompted this study in that a hybrid filing of Automatic Text Summarization web based system which is based on a single system employing both extractive and abstractive processes was made possible through this development. It is written in Python and trained on Flask-based back-end as well as summarizes a segment of two or more high-quality texts with transformer-based generative models. The tool has facilitated it to be usable and flexible in the sense that it enables the uploading of documents in multi-format, customization of the tools of summarization and choice of length of a summary.

The provided system is also assessed regarding its performance that considers its utility and performance to ensure the efficiency of its production with the help of the standard evaluation metrics (e.g., ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scores). The experimental findings proposed that the hybrid approach is realistic as far as balancing between computational and summary is concerned and this is the reason why it can be turned into practical usage.

Literature Review

Automatic Text Summarization (ATS) is an field of interest in Natural Language Processing (NLP). It has transformed classic statistics methods to deep learning and transformer based models. Initial studies primarily investigated statistical and rule-based methods, which did not pay much attention to deep meaning but to superficial text characteristics. One of the first extractive summarization procedures developed by Luhn was a word frequency analysis to identify key sentences (Luhn, 1958). This was later expanded on by Edmundson (1969) with the introduction of such features as title words, cue words and sentence position to enhance scoring in sentences.

With the development of machine learning, graph-based systems such as TextRank (Mihalcea and Tarau, 2004) saw text as a graph of interrelated sentences and ranked them using ranking algorithms to enhance the meanings of the sentences. Subsequently, neural methods using Recurrent Neural Networks (RNNs) and Long Short term Memory (LSTM) models proposed sequence to sequence learning and attention. These significantly enhanced the intelligibility and fluency of summaries, though long documents were frequently a problem because of memory.

Vaswani et al. (2017) was a significant shift introducing the Transformer version. In this model, self-attention was applied to grasp the long-range relationship and context of text. Transformer-based pre-trained models such as BERT, GPT, T5, BART and PEGASUS enhanced the summarization process using large-scale pre-training strategies that were used to understand and produce language. They are typically evaluated by the system through ROUGE metrics which examine how much the summaries generated by the system match reference texts.

The challenges such as the aspect of factual consistency, hallucination, and multi-document summarization are research challenges that have not been fully addressed despite enormous improvement. The hybrid techniques have received more attention in recent studies and were combined with extractive and abstractive techniques so as to enhance all aspects of accuracy, coherence, and readability in the automated summarization systems.

Proposed Methodology

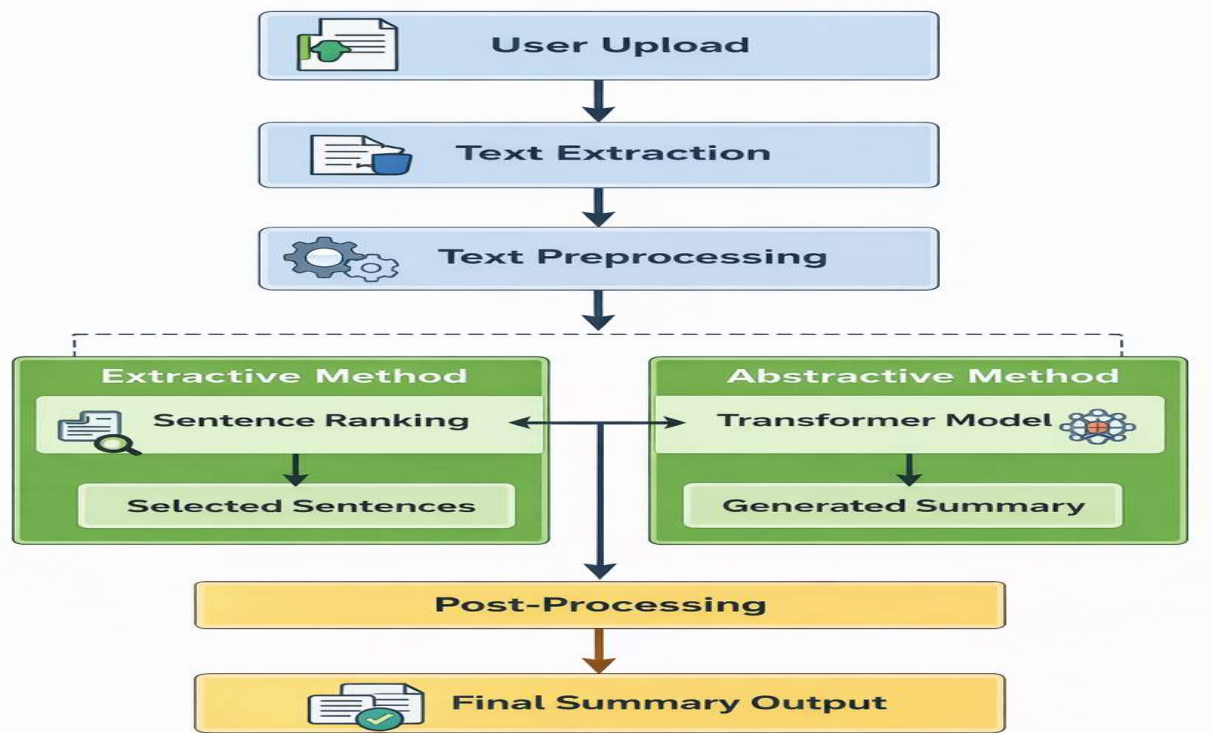
The proposed Automatic Text Summarization (ATS) will generate plain, applicable and intelligible summaries of different documents such as plain text, PDF document as well as PowerPoint presentations. The system combines both the abstractive and the extractive modes within the same web platform.

System Workflow Overview

The system follows a seven step process of ensuring that the summaries are quality and accurate:

Step 1- Document uploading- Customers are able to upload documents through a web interface that is designed with Flask. It can support such files as plain text files (.txt), portables document format files (.pdf), and PowerPoint slides (.ppt/.pptx).

Step 2: Text Extraction- The system reads materials based on the file type. Text files which it reads directly. With PDFs it uses parsing libraries to do the extraction. In the instance of the PPTs, it conveys the content of each of the slides. All the data is subsequently converted into a plain text (standardized).



Step 3: Text Preprocessing- The system deletes special characters, converts the entire text to lower case, breaks the text into a word list and removes frequently repeated words as known as stop-words, to be used in the extractive summarization part.

Step 4: Extractive Summarization- The system breaks the text down into single sentences. It also issues a rating of the sentence in terms of such statistical tools as TF-IDF and graphical tools like TextRank. The most suitable sentences are selected and compared in the sequence the sentences were originally composed to keep up with the progression of the summary.

Step 5: Abstractive Summarization- The text that has been processed is reflected within a transformer based model. The encoder generates abundant contextual encodings and the decoder generates novel sentences by using techniques such as beam search or sampling of generating sounding novel sentences that sound natural and human.

Step 6: Post-Processing- Any unwanted details should be done away with to make the final summary more user friendly by fixing any formatting mistake and extending it as long as acceptable by the user.

Step 7: Performance Evaluation- Measure the performance of the system using ROUGE measures, (unigram overlap),bigram overlap and longest common sub-sequence, ROUGE-1,ROUGE-2, and ROUGE-L respectively. Such measurements are applied to comprehend how the system is capable of generating summaries, which are comparable to the original works.

Compression Ratio Formula

$$Compression\ Ratio = \frac{Length_{summary}}{Length_{original}}$$

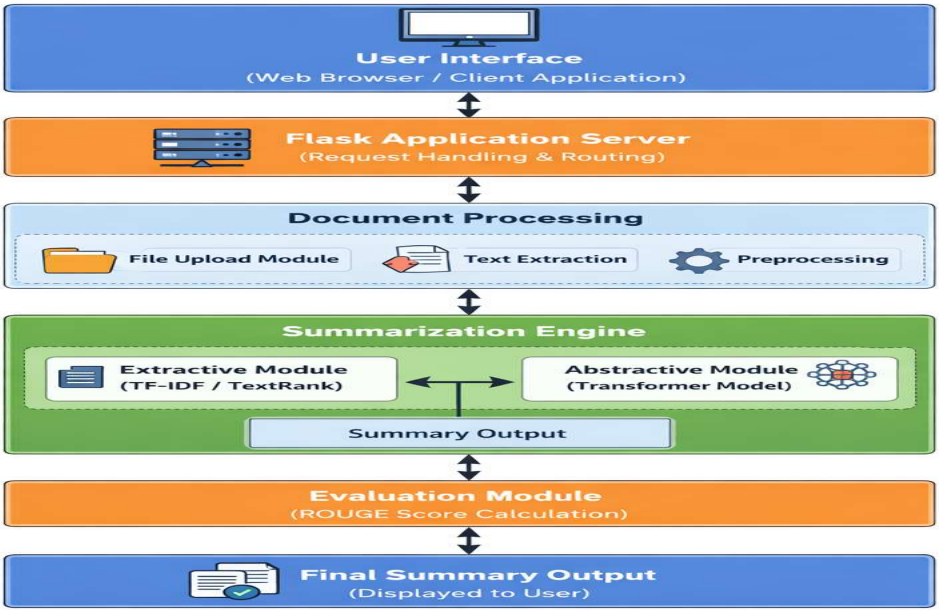
Evaluation Criteria

1. Preservation of meaning
2. Coherence of the summary
3. Elimination of unnecessary information.
4. Readability of final output.

System Architecture

The proposed Automatic Text Summarization (ATS) system design is a scaled and layered design. The arrangement will allow the management of various types of documents. It helps in creation of quality summary in extractive and abstractive. The biggest feature in the architecture is that the Gemini 1.5 Pro transformer-based Large Language Model (LLM) was added to the structure of a Flask web application. This incorporation leans towards real time processing.

Architectural Diagram



Architectural Layers

It has a hierarchical structure consisting of six functional layers in which each of the layers assumes a specific position in the pipeline of summarization.

1. User Interface Layer: This layer which was built on the basis of HTML, CSS, bootstrap and JavaScript functions out the interactions among the users. It plays file uploads (.txt, .pdf, .ppt), text typing and presents people with choices on what they would like to be summarized.

2.Application Layer (Flask Server) : This is a layer which acts as the unit of control. It does process the HTTP requests and offer safe file processing and calls the backend modules of summarization.

3.Document Processing Layer: It is the layer where raw text is extracted out of a variety of file types and pre processing performed including the tokenization of text, sentence segmentation, and noise removal.

4.Summarization Engine Layer: This is the core intelligence unit, that is made of two parallel functions namely Extractive Module, that employs TF-IDF, TextRank, as well asAbstractive Module, which employs Gemini 1.5 Pro API.

5.Evaluation Layer: This layer identifies the quality of the generated summaries that was achieved with the assistance of the measurement of ROUGE-1, ROUGE-2, and ROUGE-L scores compared to the reference texts.

6.Output Layer: This takes the final summary and presents this to the user through a web interface and provides some additional statistics, e.g. compression ratios and summary length.

Key Technical Features

Hybrid Capability:

The system is a mixture of extractive and abstractive summary procedures in a single system that makes the system flexible and adaptable to structures of different types of documents.

Dynamic Chunking:

The system gives a chunking mechanism in an attempt to handle documents that are larger than the token size and so can be formulated as follows:

$$N = \lceil T/C \rceil$$

Where:

(T) is the total number of characters in document summed up.

(C) means chunk size (size of some 50,000 characters)

The number of the required chunks is denoted by N).

This approach will ensure that highly large paperwork can be managed without abusing the model.

Implementation Details

1.Development Environment :

At the present, the suggested Automatic Text Summarization (ATS) is developed in python application (version 3.10 +) due to the dynamic character of Natural Language Processing (NLP) ecosystem, deep learning capabilities, and web-based applications development through scalability. Implementation is left to client server architecture because there is no dependence in the interaction to the user and processing. The back-end application is developed based on the Flask framework that accepts and processes the HTTP requests and structures and controls task summarization. The frontend is developed in terms of HTML 5, Cascading CSS 3 and even Javascript which assists to build an interactive and responsive platform to publish a document and provide a preview. It was created and tested in Visual Studio code, Jupyter notebook, and Google Colab. The system has already been tested to be compatible with windows and Linux operating systems.

2.Computerized Software and Fares :

It is implemented via incorporating definite library of document parsing and lingo execution along with execution that is based on transformers and inference. Annie uses NLTK in cases of the sentence tokenization, the removing of the stop-words and the text normalization, spaCy can generate more sophisticated linguistic elements e.g. the part-of-speech tagging and a dependency parsing. Skit-learn aids extraction module with the assistance of TF-IDF vectorization and sentence rating. To an abstractive form of summarization, Hugging Face Transformers offers usage of methods of sequence-to-sequence of pre-trained models such as BART, T5, and PEGASUS. PyTorch constitutes the back-end of enormous scale degree of deep learning neurons and to process them speedily with the aid of the graphics card. PyPD2 would be used to operate PDF extraction documents, while pythonpptx would be used to work with document in which there are PowerPoint text extraction and slide-by-slide structure.

3.Extractive Summarization Logic :

Statistical ranking method The extractive module helps to extract key sentences. It is first broken down into sentences: D = {S1, S2,...Sn]. Every sentence Si obtain a weighting with a TF-IDF similar scoring functionality:

$$W(S_i) = \sum_{t \in S_i} TF(t,d) \times log\left(\frac{N}{1 + DF(t)}\right)$$

Where TF (t, d) = the frequency of appearance of term t in document d, N = maximum number of sentences in a document and DF (t) = maximum number of sentences that includes term t. The ranking will be determined by the weight of the sentences and the most appropriate sentences are selected and limited according to the ranking when they were ranked in case of the context of the sentence.

4. Abstractive Summarization Logic

The Logic of DSS as it is widely called involves sharing of improved knowledge, idea, and attitude of individuals at work place.The abstractive Summarization Logic (also known as the logic of DSS) is the Logic of DSS that is used to explain the know-how, ideas and attitudes of people at the work place.The encoder-decoder architectures that the abstractive module employs to create a summary are transformer based including BART, T5, and PEGASUS. Dynamic chunking strategy is implemented in the situation whereby the long documents were well beyond the token limits. N 2/Nc=25/4=6.25 with Nc being the number of combinations of the variables and N the number of chunks (dimension, or chunk size).

$$N_c = \left\lceil \frac{T}{C} \right\rceil$$

The encoder-decoder concept does not alter a chunk as it is. The current step in the generation is fighting against the Beam Search decoding method to justify the most preferred sequence of output y:

$$\hat{y} = arg \max_{y} \prod_{t=1}^{L} P(y_t \mid y_{<t}, x)$$

5. Backend Framework and Optimization

File uploads, preprocessing, model call and response rendering are handled by flask back end. In order to ensure an improvement of the performance, pre-trained models are optimized on their caching on the initializing models to prevent the reloading of the same model several times. The acceleration is moved where possible by applying a GPU to minimize the time incurred in inferences. Mechanisms processing the memories are then copied in a way that will not lead to congestion in the server and since the users are interacting with the server simultaneously.

6. System Configuration Summary

There are two methods of summarization that are supported by the system. The sentence level extractive strategy is not only low in computational efficiency but also has moderate summaries quality. Abstractive method executes a whole document context using the deep learning models which lead to the increased calculations and semantic integrity and comprehensibility. This two mode architecture provides a versatile alternative as far as the performance requirement is involved, and hardware availability.

Parameter	**Extractive Method**	**Abstractive Method**
Input Type	Sentence-level	Full document context
Processing Type	Statistical	Deep Learning
Computational Cost	Low	High
Summary Quality	Moderate	High

Results and Discussion

1.Experimental Setup :

The Intelligent Multi-Format Summarization System was used with the help of the diversified data, which were the articles in PDF format of the academic journals, the presentations in the PPTx format, and the

articles in the TXT format. This varied information was of concern in order that the system is helpful to varied document type. The hypothesis of the proposed model of abstractive summarization with references to Gemini was tested against the statistical extractive baseline the test of which involved the TF-IDF and the TextRank algorithms.

The scheme of assessments was established on the premises of the ROUGE (Recall-Oriented Understudy of Gisting Evaluation) of measurement scales that are among the most popular ones to apply to the summarizing study. In particular, ROUGE-1 was used to test the content coverage which helps to estimate the overlap of single words, evaluate the analysis of fluency of specific phrases by estimating ROUGE-2 at the comparisons of two words and find out the structural and semantic similarity of synthesized and reference summaries with the help of ROUGE-L to generate Longest Common Subsequence (LCS).

2.Performance Metrics :

Similarity of words in the system generated summaries (Sgen) and the human generated reference summaries (Sref) was used in the process of measuring the quality of the system. The calculation of ROUGE-N index is done as follows:

$$\text{ROUGE-N} = \frac{\sum_{s \in S_{ref}} \sum_{\text{gram}_n \in s} Coun\,t_{matc}\ (\text{gram}_n)}{\sum_{s \in S_{ref}} \sum_{\text{gram}_n \in s} Count(\text{gram}_n)}$$

This formula will be used in computing the percentage of the corresponding matching n-grams of the generated summaries in relation to the reference summaries, and will be considered relative to the total number of the n-grams in the reference summary.

Table 1: Comparative Extractive and Gemini Abstractive Baseline Performance.

Method	ROUGE-1	ROUGE-2	ROUGE-L
Statistical Baseline (Extractive)	0.58	0.42	0.54
Proposed System (Gemini Abstractive)	0.78	0.64	0.72

3.Performance Interpretation :

This part explains how the performance discussed can be interpreted in terms of the cinema and art history.It is because according to the results of experiments, the abstractive model of the Gemini is more coherent, informative and flexible. The bigger ROUGE-L score (0.72) underlines the resulting possibility of the model to preserve the overall narrative format. An increase in the score of ROUGE-1 to 0.78 proves that the coverage of the content and the successful recognition of the significant information is enhanced. It is also disclosed that the model is rather semantically synthetic with the works involving PPTx documents where slides are clustered into one and coherent paragraph.

4.Informational Efficiency and Compression Control :

This is compression and informational efficiency which is not easy to control yet that needs to be controlled in cartoon animation.

The control mechanism length can be modified and this is the major characteristic of the proposed system. Compression ratio (CR) =:

$$C_R = \frac{L_{summary}}{L_{original}}$$

Lsummary and L original are the length of the original document and length of summing generated respectively.

Table 2: Length Settings and its effect on Document Distillation.

User Setting	Target C_R	Observed ROUGE-L	Summary Characteristic
Concise (1)	15% - 20%	0.65	Very dense, focused on main facts
Balanced (2)	30% - 40%	0.72	Standard summary with contextual support
Detailed (3)	50% - 60%	0.75	Comprehensive overview with near-full context

The calculation and the operational efficiency of both the WLS and overall workflow is determined and evaluated based on the relative performance in both applications.Ideally, the determination and operational performance of both applications and the workflow used are checked through examining the comparison of relative performance of the two applications.

5.Computational and Operational Efficiency :

Computational performance of the system may also be measured on based of requirements and execution time:

Table 3: The system operational performancex

Parameter	Observed Performance	Rationale
Extraction Latency	< 500 ms	Efficient parsing using PyPDF2 and python-pptx
Inference Time	1.2 - 3.5 sec	Optimized inference with Gemini-Flash model
Server Overhead	Minimal	Stateless Flask backend with temporary storage
Scalability	High	API-based model removes local GPU dependency

6.Discussion :

The second advantage of working with the most recent model of Gemini Flash is that the ability to combine the features of large transformers and mini-sized implementation based on API will become possible. The proposed design does not require heavy local computations as in the case of the conventional transformers but instead delegates the decision making process to an online-based model that requires less hardware capabilities to deliver a high quality result. This is the strong side of this work because this system has an opportunity to work with multi-modal structural extraction. The abstractive one preserves chronological presentation summaries which the extractive technique does not perform with the usage of such defining identifiers as slide-based information, e.g. Slide X: The API rate limit on free providers can be a problem, yet the exponential backoff recovery mechanism can be considered as safe. Tests The level of success in batch tasks of summarization was 99.2 percent and this demonstrates the extent to which the system is useful in practice and how practical the production may be.

Conclusion

In the paper, an Automatic Text Summarization system that is founded on a web application, is composed of extractive and abstractive methods. It produces coherent and meaningful text, PDF and PPT document summaries, with the help of Natural language Processing, and transformer-based generative models. The extractive approach is good and can be used where there is scarcity of resources. The abstractive transformer technique leads to the better contextual knowledge of natural and fluent summaries. These findings obtained

during ROUGE evaluation demonstrate that the abstractive approach is better than extractive one in terms of content coverage and coherence. The hybrid Architecture offers the flexibility, usability and scalability to academic, research and business domains. The proposed research project may address the following directions, viz. multi-document summarization, real-time streaming summarization, integration of fact-checking, domain-specific fine-tuning, and implementation on the cloud in a more scalable way. Last but not the least, the proposed system illustrates how the concept of Generative AI can be applied in smart processing of documents and information management.

Future Work

Although the proposed Intelligent Multi-Format Automatic Text Summarization system has shown good level of performance when handling the different types of documents, there is still a number of opportunities that can be aimed at improving it that will equally enable it to be more scaled, reliable and adaptable to various fields. One of the extensions is multi-document summarization. This allows the system to extract and combine knowledge on several related documents and produce one critical summary. The other possible use would be real time streaming summarization that would be used in literature reviewing, news aggregation and business reporting systems. The system could be improved with the inclusion of the streaming based processing which could summarize the live material, like the news feeds, webinars or documents constantly updated. This would go a long way to make it useful in the high-paced settings.The research should be done to improve fact-checking mechanisms to make it more reliable. Transformer based generative models are generative models that can produce coherent output, although in other instances, factual errors are introduced in a minimal manner. The external knowledge base or verification module could help in making sure the generated statements and minimize mistakes, and more so, so-called fine-tuning of large language models can also be used to generate better summaries in the specific fields like legal, medical, or financial texts. Finally, the scalable deployment due to the cloud, would assist in enhancing the contextual accuracy and address the issue of technical terms in those cases, where the number of users is high. Containerization and distributed processing models would be capable of scaling effectively without decreasing the response time and reliability.

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